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9.5 Mechanism of muscle contraction

Sliding filament theory in 1954, Andrew F. Huxley and Rolf Niedergerke proposed the sliding-filament theory to explain muscle contraction. According to this theory, overlapping actin and myosin filaments of fixed length slide past one another in an energy requiring process, resulting in muscle contraction. The contraction of muscle fibre is a remarkable process that helps in creating a force to move or to resist a load. The force which is created by the contracting muscle is called muscle tension. The load is a weight or force that opposes contraction of a muscle. Contraction is the creation of tension in the muscle which is an active process and relaxation is the release of tension created by contraction. Muscle contraction is initiated by a nerve impulse sents by the

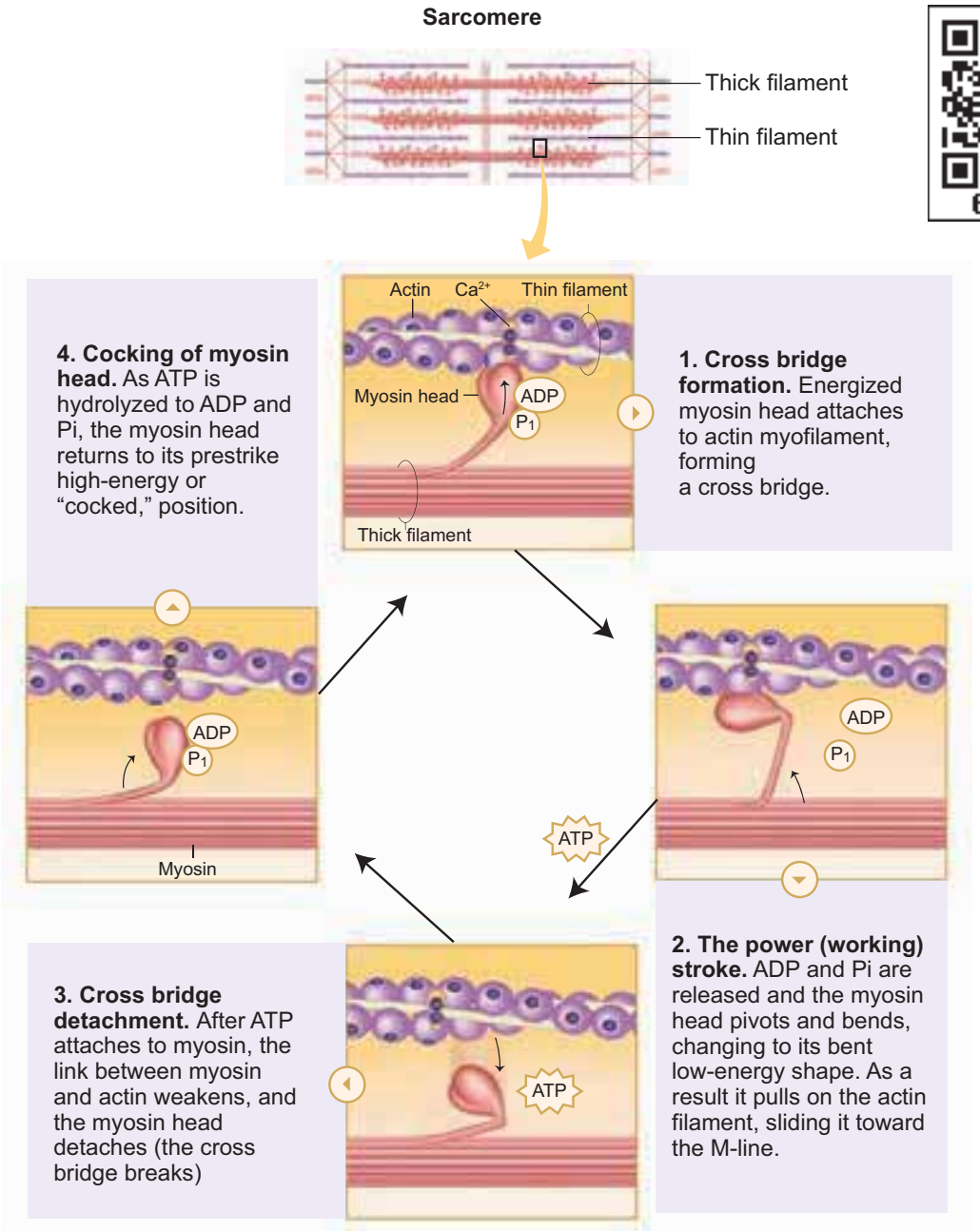


Figure 9.4 Cross-bridge cycle of muscle contraction